

NCERT Solutions For Class 12 Physics Chapter 11 Dual Nature Of Radiation And Matter

Q.1 Find the following:

(a) Maximum frequency, and

(b) The minimum wavelength of X-rays produced by 30 kV electrons.

Ans:

To find the maximum frequency and the minimum wavelength of X-rays, we use energy formulas related to electron acceleration and Planck's constant.

- Given: $V = 30 \text{ kV} = 3 \times 10^4 \text{ V}$; Electron energy $E = 3 \times 10^4 \text{ eV}$; $e = 1.6 \times 10^{-19} \text{ C}$; $h = 6.626 \times 10^{-34} \text{ Js}$.
- (a) Energy of electrons: $E = h\nu \Rightarrow \nu = E/h = (1.6 \times 10^{-19} \times 3 \times 10^4)/(6.626 \times 10^{-34}) = 7.24 \times 10^{18} \text{ Hz}$.
- Maximum frequency of X-rays $= 7.24 \times 10^{18} \text{ Hz}$.
- (b) Minimum wavelength: $\lambda = c/\nu = (3 \times 10^8)/(7.24 \times 10^{18}) = 4.14 \times 10^{-11} \text{ m} = 0.0414 \text{ nm}$.

Q.2 The work function of cesium metal is 2.14 eV. When the light of frequency $6 \times 10^{14} \text{ Hz}$ is incident on the metal surface, photoemission of electrons occurs. Find the following:

(a) The maximum kinetic energy of the emitted electrons

(b) Stopping potential

(c) The maximum speed of the emitted photoelectrons

Ans:

We find the KE, stopping potential, and max speed using the photoelectric effect equations with the given work function and incident frequency.

- Work function $\Phi_0 = 2.14 \text{ eV}$; Frequency $\nu = 6.0 \times 10^{14} \text{ Hz}$; $h = 6.626 \times 10^{-34} \text{ Js}$.
- (a) Max kinetic energy: $K = h\nu - \Phi_0 = (6.626 \times 10^{-34} \times 6 \times 10^{14}) / (1.6 \times 10^{-19}) - 2.14 = 2.485 - 2.14 = 0.345 \text{ eV}$.
- (b) Stopping potential: $K = eV_0 \Rightarrow V_0 = K/e = 0.345 \text{ V}$.
- (c) Max speed: $K = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2K/m} = \sqrt{2 \times 0.345 \times 1.6 \times 10^{-19} / 9.1 \times 10^{-31}} = 3.323 \times 10^5 \text{ m/s} = 332.3 \text{ km/s}$.

Q.3 The photoelectric cut-off voltage in a certain experiment is 1.5 V. What is the maximum kinetic energy of photoelectrons emitted?

Ans:

Photoelectric cut-off voltage, $V_0 = 1.5 \text{ V}$.

For emitted photoelectrons, the maximum kinetic energy is:

$$K_e = eV_0 = 1.6 \times 10^{-19} \times 1.5 = 2.4 \times 10^{-19} \text{ J}$$

Therefore, $2.4 \times 10^{-19} \text{ J}$ is the maximum kinetic energy emitted by the photoelectrons.

Q.4 Monochromatic light of wavelength 632.8 nm is produced by a helium–neon laser. The power emitted is 9.42 mW.

- (a) Find the energy and momentum of each photon in the light beam.**
- (b) How many photons per second, on average, arrive at a target irradiated by this beam? (Assume the beam to have a uniform cross-section which is less than the target area)**
- (c) How fast does a hydrogen atom have to travel in order to have the same momentum as that of the photon?**

Ans:

We calculate the energy, momentum, photon rate, and the comparable atom speed stepwise using standard photon formulas.

- $\lambda = 632.8 \text{ nm} = 632.8 \times 10^{-9} \text{ m}$; $P = 9.42 \times 10^{-3} \text{ W}$; $h = 6.626 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ m/s}$; $m (\text{H atom}) = 1.66 \times 10^{-27} \text{ kg}$.
- (a) Energy of photon: $E = hc/\lambda = 3.141 \times 10^{-19} \text{ J}$. Momentum: $P = h/\lambda = 1.047 \times 10^{-27} \text{ kg m/s}$.
- (b) Number of photons/sec: $n = P/E = 9.42 \times 10^{-3} / 3.141 \times 10^{-19} = 3 \times 10^{16} \text{ photons/s}$
- (c) Hydrogen atom speed for same momentum: $v = P/m = 1.047 \times 10^{-27} / 1.66 \times 10^{-27} = 0.621 \text{ m/s}$.

Q.5 The energy flux of sunlight reaching the surface of the earth is $1.388 \times 10^3 \text{ W/m}^2$. How many photons are incident on the earth per second/square metre? Assume an average wavelength of 550 nm.

Ans:

By dividing the power per square metre by energy per photon, we get the number of incident photons.

- Energy flux $\phi = 1.388 \times 10^3 \text{ W/m}^2$; $\lambda = 550 \text{ nm} = 550 \times 10^{-9} \text{ m}$; $h = 6.626 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ m/s}$.
- Energy per photon: $E = hc/\lambda$
- Number of photons per metre² per second: $n = P/E = (P\lambda)/(hc) = (1.388 \times 10^3 \times 550 \times 10^{-9}) / (6.626 \times 10^{-34} \times 3 \times 10^8) = 3.84 \times 10^{21} \text{ photons/m}^2/\text{s}$.
- So, 3.84×10^{21} photons hit each square metre every second.

Q.6 In an experiment on the photoelectric effect, the slope of the cut-off voltage versus frequency of incident light is found to be $4.12 \times 10^{-15} \text{ V s}$. Calculate the value of Planck's constant.

Ans:

Using the slope of the cut-off voltage-frequency graph and the charge of the electron, we can find Planck's constant.

- Slope $(\Delta V/\Delta \nu) = 4.12 \times 10^{-15} \text{ V}\cdot\text{s}$
- $e = 1.6 \times 10^{-19} \text{ C}$
- Planck's constant: $h = e \times \text{slope} = 1.6 \times 10^{-19} \times 4.12 \times 10^{-15} = 6.592 \times 10^{-34} \text{ Js}$.

Q.7 A 100 W sodium lamp radiates energy uniformly in all directions. The lamp is located at the centre of a large sphere that absorbs all the sodium light which is incident on it. The wavelength of sodium light is 589 nm.

(a) What is the energy per photon associated with sodium light?

(b) At what rate are the photons delivered to the sphere?

Ans:

We find energy per photon using the wavelength and calculate photons per second using the power of the lamp.

- Power $P = 100 \text{ W}$; $\lambda = 589 \text{ nm} = 589 \times 10^{-9} \text{ m}$; $h = 6.626 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ m/s}$.
- (a) $E = hc/\lambda = (6.626 \times 10^{-34} \times 3 \times 10^8) / (589 \times 10^{-9}) = 3.37 \times 10^{-19} \text{ J} = 2.11 \text{ eV}$.
- (b) Number of photons/sec: $n = P/E = 100 / 3.37 \times 10^{-19} = 2.96 \times 10^{20} \text{ photons/s}$.

Q.8 The threshold frequency for a certain metal is $3.3 \times 10^{14} \text{ Hz}$. If the light of frequency $8.2 \times 10^{14} \text{ Hz}$ is incident on the metal, predict the cut-off voltage for the photoelectric emission.

Ans:

Use the difference of the frequencies and constants to calculate the cut-off voltage for the metal.

- $\nu_0 = 3.3 \times 10^{14} \text{ Hz}$; $\nu = 8.2 \times 10^{14} \text{ Hz}$; $e = 1.6 \times 10^{-19} \text{ C}$; $h = 6.626 \times 10^{-34} \text{ Js}$.
- Cut-off voltage: $V_0 = h(\nu - \nu_0)/e = [6.626 \times 10^{-34} \times (8.2 \times 10^{14} - 3.3 \times 10^{14})] / 1.6 \times 10^{-19} = 2.0291 \text{ V}$.

Q.9 The work function for a certain metal is 4.2 eV. Will this metal give photoelectric emission for incident radiation of wavelength 330 nm?

Ans:

Work function $\Phi_0 = 4.2 \text{ eV}$.

Wavelength $\lambda = 330 \text{ nm} = 330 \times 10^{-9} \text{ m}$.

Energy per photon: $E = hc/\lambda = 6.0 \times 10^{-19} \text{ J} = 3.76 \text{ eV}$.

Since the photon energy is less than the work function ($3.76 \text{ eV} < 4.2 \text{ eV}$), there will be no photoelectric emission.

Q.10 Light of frequency $7.21 \times 10^{14} \text{ Hz}$ is incident in a metal surface. Electrons with a maximum speed of $6.0 \times 10^5 \text{ m/s}$ are ejected from the surface. What is the threshold frequency for the photoemission of electrons?

Ans:

The threshold frequency is found by subtracting the velocity term from incident frequency using the kinetic energy relation.

- $v(\text{max}) = 6.0 \times 10^5 \text{ m/s}$; $v = 7.21 \times 10^{14} \text{ Hz}$; $h = 6.626 \times 10^{-34} \text{ Js}$; $m = 9.1 \times 10^{-31} \text{ kg}$.
- Threshold frequency: $v_o = v - (mv^2/2h) = 7.21 \times 10^{14} - [(9.1 \times 10^{-31}) \times (6 \times 10^5)^2 / (2 \times 6.626 \times 10^{-34})] = 4.738 \times 10^{14} \text{ Hz}$.

Q.11 Light of wavelength 488 nm is produced by an argon laser which is used in the photoelectric effect. When light from this spectral line is incident on the emitter, the stopping (cut-off) potential of photoelectrons is 0.38 V. Find the work function of the material from which the emitter is made.

Ans:

The work function is found by subtracting the stopping potential energy from the photon's energy in eV.

- $\lambda = 488 \text{ nm} = 488 \times 10^{-9} \text{ m}$; $V_o = 0.38 \text{ V}$; $e = 1.6 \times 10^{-19} \text{ C}$; $h = 6.6 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ m/s}$.
- Work function: $\Phi_o = (hc/\lambda) - eV_o = [(6.6 \times 10^{-34} \times 3 \times 10^8)/(1.6 \times 10^{-19} \times 488 \times 10^{-9})] - 0.38 = 2.54 - 0.38 = 2.16 \text{ eV}$.

Q.12 Calculate the following:

(a) Momentum

(b) The de Broglie wavelength of the electrons accelerated through a potential difference of 56 V.

Ans:

We use the kinetic energy and de Broglie relations to find the momentum and wavelength of electrons.

- $V = 56 \text{ V}$; $h = 6.6 \times 10^{-34} \text{ Js}$; $m = 9.1 \times 10^{-31} \text{ Kg}$; $e = 1.6 \times 10^{-19} \text{ C}$.
- (a) Electron speed $v = \sqrt{2eV/m} = \sqrt{2 \times 1.6 \times 10^{-19} \times 56 / 9.1 \times 10^{-31}} = 4.44 \times 10^6 \text{ m/s}$
- Momentum $p = m \times v = 9.1 \times 10^{-31} \times 4.44 \times 10^6 = 4.04 \times 10^{-24} \text{ kg m/s}$
- (b) de Broglie wavelength $\lambda = 12.27/\sqrt{V} \text{ \AA} = 12.27/\sqrt{56} = 0.1639 \text{ nm}$

Q.13 Find the following:

(a) Momentum,

(b) Speed, and

(c) De Broglie wavelength of an electron with a kinetic energy of 120 eV.

Ans:

Calculate each value step wise, starting with speed from energy, then using the results to get momentum and wavelength.

- Electron KE (E_k) = $120 \text{ eV} = 120 \times 1.6 \times 10^{-19} \text{ J}$
- $m = 9.1 \times 10^{-31} \text{ Kg}$; $h = 6.6 \times 10^{-34} \text{ Js}$.
- Speed $v = \sqrt{2eE_k/m} = \sqrt{2 \times 1.6 \times 10^{-19} \times 120 / 9.1 \times 10^{-31}} = 6.496 \times 10^6 \text{ m/s}$

- Momentum $p = m \times v = 9.1 \times 10^{-31} \times 6.496 \times 10^6 = 5.91 \times 10^{-24} \text{ kg m/s}$
- De Broglie wavelength $\lambda = h/p = 6.6 \times 10^{-34} / 5.91 \times 10^{-24} = 1.116 \times 10^{-10} \text{ m}$
= 0.112 nm

Q.14 The wavelength of light from the spectral emission line of sodium is 589 nm. Find the kinetic energy at which
(a) an electron, and
(b) a neutron, would have the same de Broglie wavelength.

Ans:

The KE for electron and neutron is calculated using the de Broglie relation for the same wavelength.

- $\lambda = 589 \text{ nm} = 589 \times 10^{-9} \text{ m}$; $m_e = 9.1 \times 10^{-31} \text{ Kg}$; $m_n = 1.66 \times 10^{-27} \text{ Kg}$; $h = 6.6 \times 10^{-34} \text{ Js}$.
- (a) Electron $K = h^2 / (2\lambda^2 m_e) = (6.6 \times 10^{-34})^2 / (2 \times (589 \times 10^{-9})^2 \times 9.1 \times 10^{-31}) = 6.9 \times 10^{-25} \text{ J} = 4.31 \times 10^{-6} \text{ eV}$
- (b) Neutron $K = h^2 / (2\lambda^2 m_n) = (6.6 \times 10^{-34})^2 / (2 \times (589 \times 10^{-9})^2 \times 1.66 \times 10^{-27}) = 3.78 \times 10^{-28} \text{ J} = 2.36 \times 10^{-9} \text{ eV} = 2.36 \text{ neV}$

Q.15 What is the de Broglie wavelength of:

- (a) a bullet of mass 0.040 kg travelling at the speed of 1.0 km/s,**
- (b) a ball of mass 0.060 kg moving at a speed of 1.0 m/s, and**
- (c) a dust particle of mass $1.0 \times 10^{-9} \text{ kg}$ drifting with a speed of 2.2 m/s?**

Ans:

We use the de Broglie formula $\lambda = h/mv$ for each case.

(a) Bullet: $m = 0.040 \text{ kg}$, $v = 1000 \text{ m/s}$; $\lambda = 6.6 \times 10^{-34} / (0.040 \times 1000) = 1.65 \times 10^{-35} \text{ m}$

(b) Ball: $m = 0.060 \text{ kg}$, $v = 1 \text{ m/s}$; $\lambda = 6.6 \times 10^{-34} / (0.060 \times 1) = 1.1 \times 10^{-32} \text{ m}$

(c) Dust: $m = 1 \times 10^{-9} \text{ kg}$, $v = 2.2 \text{ m/s}$; $\lambda = 6.6 \times 10^{-34} / (2.2 \times 10^{-9}) = 3.0 \times 10^{-25} \text{ m}$

Q.16 An electron and a photon each have a wavelength of 1.00 nm. Find the following:

(a) Momenta

(b) The energy of the photon

(c) The kinetic energy of the electron

Ans:

Equal wavelengths mean equal momenta; then, energies are calculated with appropriate equations for photon and electron.

$$\lambda = 1 \text{ nm} = 1 \times 10^{-9} \text{ m}; h = 6.63 \times 10^{-34} \text{ Js}$$

(a) Both momenta: $p = h/\lambda = 6.63 \times 10^{-34} / 1 \times 10^{-9} = 6.63 \times 10^{-25} \text{ kg m/s}$

(b) Photon energy: $E = hc/\lambda = (6.63 \times 10^{-34} \times 3 \times 10^8) / (1 \times 10^{-9} \times 1.6 \times 10^{-19}) = 1243.1 \text{ eV} = 1.243 \text{ keV}$

(c) Electron kinetic energy: $K = (1/2)p^2/m = (1/2) \times (6.63 \times 10^{-25})^2 / (9.1 \times 10^{-31}) = 2.415 \times 10^{-19} \text{ J} = 1.51 \text{ eV}$

Q.17 (a) For what kinetic energy of a neutron will the associated de Broglie wavelength be $1.40 \times 10^{-10} \text{ m}$?

(b) Also find the de Broglie wavelength of a neutron in thermal equilibrium with matter, having an average kinetic energy of $(3/2) kT$ at 300 K.

Ans:

Relate Kinetic Energy and wavelength via the de Broglie relation, then substitute for thermal equilibrium at room temperature.

(a) $\lambda = 1.40 \times 10^{-10} \text{ m}$; $m_n = 1.66 \times 10^{-27} \text{ Kg}$; $h = 6.63 \times 10^{-34} \text{ Js}$

$$K = h^2 / (2\lambda^2 m_n) = (6.63 \times 10^{-34})^2 / (2 \times (1.40 \times 10^{-10})^2 \times 1.66 \times 10^{-27}) = 6.75 \times 10^{-21} \text{ J} = 4.219 \times 10^{-2} \text{ eV}$$

(b) $T = 300 \text{ K}$; $k = 1.38 \times 10^{-23} \text{ J/K}$; $K' = (3/2)kT = 6.21 \times 10^{-21} \text{ J}$

$$\lambda' = h / \sqrt{(2K'm_n)} = 1.46 \times 10^{-10} \text{ m} = 0.146 \text{ nm}$$

Q.18 Show that the wavelength of electromagnetic radiation is equal to the de Broglie wavelength of its quantum (photon).

Ans:

The momentum of a photon with energy ($h\nu$) is $p = h\nu/c = h/\lambda$. The de Broglie wavelength is defined by $\lambda = h/p$ for any particle. For a photon, $p = mv$ and $v = c$, so these two wavelengths are the same.

Q.19 What is the de Broglie wavelength of a nitrogen molecule in air at 300 K? Assume that the molecule is moving with the root-mean-square speed of molecules at this temperature. (Atomic mass of nitrogen = 14.0076 u)

Ans:

We use the rms speed for nitrogen at room temperature and apply the de Broglie equation.

- $T = 300 \text{ K}$; nitrogen molecule mass $m = 28.0152 \times 1.66 \times 10^{-27} \text{ kg}$; $h = 6.63 \times 10^{-34} \text{ Js}$; $k = 1.38 \times 10^{-23} \text{ J/K}$
- $V_{\text{rms}} = \sqrt{3kT/m}$; $\lambda = h/\sqrt{3mkT} = 0.028 \times 10^{-9} \text{ m} = 0.028 \text{ nm}$

Q.20 (a) Estimate the speed with which electrons emitted from a heated emitter of an evacuated tube impinge on the collector maintained at a potential difference of 500 V with respect to the emitter. Ignore the low initial speeds of the electrons. The specific charge of the electron, i.e. its e/m , is given to be $1.76 \times 10^{11} \text{ C kg}^{-1}$. (b) Use the same formula you employ in (a) to obtain electron speed for a collector potential of 10 MV. Do you see what is wrong? In what way is the formula to be modified?

Ans:

For the lower voltage, use non-relativistic equations; at high voltage, the velocity exceeds light speed, showing formula limits.

- (a) $V = 500 \text{ V}$, $e/m = 1.76 \times 10^{11} \text{ C kg}^{-1}$
- Speed $v = \sqrt{2Ve/m} = \sqrt{2 \times 500 \times 1.76 \times 10^{11}} = 1.32 \times 10^7 \text{ m/s}$
- (b) For $V = 10 \text{ MV}$, $v = \sqrt{2 \times 10^7 \times 1.76 \times 10^{11}} = 1.88 \times 10^9 \text{ m/s}$ (greater than c , not possible) – so the formula only works when $v \ll c$. For higher speeds, use relativistic formulas: $m = m_0/\sqrt{1-v^2/c^2}$, $K = mc^2 - m_0c^2$.

Q. 21 (a) A monoenergetic electron beam with an electron speed of $5.20 \times 10^6 \text{ m s}^{-1}$ is subject to a magnetic field of $1.30 \times 10^{-4} \text{ T}$ normal to the beam velocity. What is the radius of the circle traced by the beam, given e/m for electron equals $1.76 \times 10^{11} \text{ C kg}^{-1}$? (b) Is the formula you employ in (a) valid for calculating the radius of the path of a 20 MeV electron beam? If not, in what way is it modified?

Ans:

The radius for low velocity is easily found; for highly energetic beams, relativistic effects must be considered.

- (a) $v = 5.20 \times 10^6 \text{ m/s}$; $B = 1.30 \times 10^{-4} \text{ T}$; $e/m = 1.76 \times 10^{11} \text{ C kg}^{-1}$
- Radius $r = v/[(e/m)B] = 5.20 \times 10^6 / (1.76 \times 10^{11} \times 1.30 \times 10^{-4}) = 0.227 \text{ m} = 22.7 \text{ cm}$.

- (b) At higher energy (e.g. 20 MeV) the calculated speed exceeds c , so this formula is no longer valid; use relativistic mass $m = m_0 / \sqrt{1-v^2/c^2}$. The radius becomes $r = m v / eB$, and m must use the relativistic value.

Q.22 An electron gun with its collector at a potential of 100 V fires out electrons in a spherical bulb containing hydrogen gas at low pressure ($\sim 10^{-2}$ mm of Hg). A magnetic field of 2.83×10^{-4} T curves the path of the electrons in a circular orbit of radius 12.0 cm. (The path can be viewed because the gas ions in the path focus the beam by attracting electrons, and emitting light by electron capture; this method is known as the 'fine beam tube' method.) Determine e/m from the data.

Ans:

Combine the centripetal force and kinetic energy formulas to solve for e/m using the experiment's given values.

- $V = 100$ V; $B = 2.83 \times 10^{-4}$ T; $r = 12 \times 10^{-2}$ m
- Combine: $evB = mv^2/r$, and $(1/2)mv^2 = eV$; solve for e/m
- $e/m = 2V/(B^2r^2) = 2 \times 100 / (2.83 \times 10^{-4})^2 \times (12 \times 10^{-2})^2 = 1.73 \times 10^{11}$ C kg $^{-1}$

Q.23 (a) An X-ray tube produces a continuous spectrum of radiation with its short wavelength ending at $0.45 \text{ \AA}$. What is the maximum energy of a photon in radiation? (b) From your answer to (a), guess what order of accelerating voltage (for electrons) is required in such a tube?

Ans:

Use the photon energy-wavelength formula, then relate that energy to expected accelerating voltage.

- (a) $\lambda = 0.45 \text{ \AA} = 0.45 \times 10^{-10} \text{ m}$; $E_{\text{max}} = hc/\lambda = (6.626 \times 10^{-34} \times 3 \times 10^8) / (0.45 \times 10^{-10} \times 1.6 \times 10^{-19}) = 27.6 \text{ keV}$
- (b) The needed voltage is about 30 kV, matching 27.6 keV electron energy.

Q.24 In an accelerator experiment on high-energy collisions of electrons with positrons, a certain event is interpreted as the annihilation of an electron-positron pair of total energy 10.2 BeV into two γ -rays of equal energy. What is the wavelength associated with each γ -ray? (1BeV = 10^9 eV)

Ans:

Divide total energy equally into two photons, then apply the energy-wavelength relation.

- Total $E = 10.2 \times 10^9 \text{ eV}$; each γ : $E' = 5.1 \times 10^9 \text{ eV} = 8.16 \times 10^{-10} \text{ J}$
- $\lambda = hc/E' = (6.626 \times 10^{-34} \times 3 \times 10^8) / 8.16 \times 10^{-10} = 2.436 \times 10^{-16} \text{ m}$

Q.25 Estimating the following two numbers should be interesting. The first number will tell you why radio engineers do not need to worry much

about photons. The second number tells you why our eye can never 'count photons', even in barely detectable light.

(a) The number of photons emitted per second by a medium wave transmitter of 10 kW power, emitting radio waves of wavelength 500 m.

(b) The number of photons entering the pupil of our eye per second corresponding to the minimum intensity of white light that we humans can perceive ($\sim 10^{-10} \text{ W m}^{-2}$). Take the area of the pupil to be about 0.4 cm^2 , and the average frequency of white light to be about $6 \times 10^{14} \text{ Hz}$.

Ans:

For (a), divide power by photon energy. For (b), calculate photon flux, then multiply by the area of the pupil.

- (a) Power: 10^4 W ; $\lambda = 500 \text{ m}$; photon energy $E' = hc/\lambda = 3.96 \times 10^{-28} \text{ J}$
- Number/sec: $n = 10^4 / 3.96 \times 10^{-28} \approx 2.5 \times 10^{31}$
- (b) Intensity: 10^{-10} W/m^2 ; area = $0.4 \times 10^{-4} \text{ m}^2$; $\nu = 6 \times 10^{14} \text{ Hz}$
- Photon energy $E = h\nu = 3.96 \times 10^{-19} \text{ J}$; photon flux $n = I/E = 2.52 \times 10^8 \text{ photons/m}^2/\text{s}$
- Total entering the pupil: $n \times \text{area} = 1.008 \times 10^4 \text{ s}^{-1}$

Q.26 Ultraviolet light of wavelength $2271 \text{ \AA}$ from a 100 W mercury source irradiates a photo-cell made of molybdenum metal. If the stopping potential is -1.3 V , estimate the work function of the metal. How would

the photo-cell respond to high-intensity ($\sim 10^5 \text{ W m}^{-2}$) red light of wavelength $6328 \text{ \AA}$ produced by a He-Ne laser?

Ans:

Calculate work function using UV input, then compare red light frequency with molybdenum's threshold to check for photoelectric effect.

- $\lambda = 2271 \text{ \AA} = 2271 \times 10^{-10} \text{ m}$; $V_0 = 1.3 \text{ V}$; $h = 6.6 \times 10^{-34} \text{ J}$; $e = 1.6 \times 10^{-19} \text{ C}$
- Work function $\Phi_0 = (hc/\lambda) - eV_0 = 8.72 \times 10^{-19} - 2.08 \times 10^{-19} = 6.64 \times 10^{-19} \text{ J}$
- Threshold frequency $\nu_0 = \Phi_0/h = 1.006 \times 10^{15} \text{ Hz}$; red light frequency $\nu_r = 4.74 \times 10^{14} \text{ Hz}$
- Since $\nu_r < \nu_0$, no photoelectric effect happens with red light, no matter the intensity.

Q.27 Monochromatic radiation of wavelength 640.2 nm ($1 \text{ nm} = 10^{-9} \text{ m}$) from a neon lamp irradiates photosensitive material made of caesium on tungsten. The stopping voltage is measured to be 0.54 V . The source is replaced by an iron source, and its 427.2 nm line irradiates the same photo-cell. Predict the new stopping voltage.

Ans:

Work out work function from the first source, then use it to solve for the stopping potential of the second (iron) source.

- 1st: $\lambda = 640.2 \text{ nm} = 640.2 \times 10^{-9} \text{ m}$; $V_0 = 0.54 \text{ V}$
- Work function $\Phi_0 = (hc/\lambda) - eV_0 = 3.093 \times 10^{-19} - 0.864 \times 10^{-19} = 2.229 \times 10^{-19} \text{ J}$
- 2nd: $\lambda' = 427.2 \text{ nm} = 427.2 \times 10^{-9} \text{ m}$; $eV_0' = (hc/\lambda') - \Phi_0$
- $eV_0' = 4.63 \times 10^{-19} - 2.229 \times 10^{-19} = 2.401 \times 10^{-19} \text{ J}$; $V_0' = 2.401 \times 10^{-19} / 1.6 \times 10^{-19} = 1.5 \text{ V}$

Q.28 A mercury lamp is a convenient source for studying the frequency dependence of photoelectric emission since it gives a number of spectral lines ranging from the UV to the red end of the visible spectrum. In our experiment with rubidium photo-cell, the following lines from a mercury source were used: $\lambda_1 = 3650 \text{ \AA}$, $\lambda_2 = 4047 \text{ \AA}$, $\lambda_3 = 4358 \text{ \AA}$, $\lambda_4 = 5461 \text{ \AA}$, $\lambda_5 = 6907 \text{ \AA}$. The stopping voltages, respectively, were measured to be: $V_{01} = 1.28 \text{ V}$, $V_{02} = 0.95 \text{ V}$, $V_{03} = 0.74 \text{ V}$, $V_{04} = 0.16 \text{ V}$, $V_{05} = 0 \text{ V}$. Determine the value of Planck's constant h , the threshold frequency and work function for the material.

Ans:

By plotting V_0 versus frequency and calculating the best-fit slope, h , threshold frequency and work function can be found.

- First, calculate frequencies for each line: $\nu_1 = 8.219 \times 10^{14} \text{ Hz}$, $\nu_2 = 7.412 \times 10^{14} \text{ Hz}$, etc.

- The stopping potential V_0 graph's slope: $\Delta V/\Delta \nu = (1.28 - 0.16)/(8.214 - 5.493) \times 10^{14} = 1.12/(2.726) \times 10^{14} = h/e$
- Solve for h : $h = (1.12 \times 1.6 \times 10^{-19})/(2.726 \times 10^{14}) = 6.573 \times 10^{-34} \text{ Js}$
- Threshold frequency: intersection of graph at $\nu_0 \approx 5 \times 10^{14} \text{ Hz}$
- Work function: $\Phi_0 = h\nu_0 = 6.573 \times 10^{-34} \times 5 \times 10^{14} = 3.286 \times 10^{-19} \text{ J} = 2.054 \text{ eV}$

Q.29 The work function for the following metals is given: Na: 2.75 eV, K: 2.30 eV, Mo: 4.17 eV, Ni: 5.15 eV. Which of these metals will not give photoelectric emission for radiation of wavelength 3300 Å from a He-Cd laser placed 1 m away from the photocell? What happens if the laser is brought nearer and placed 50 cm away?

Ans:

Radiation energy $E = hc/\lambda = (6.63 \times 10^{-34} \times 3 \times 10^8) / (3300 \times 10^{-10}) = 6.018 \times 10^{-19} \text{ J} = 3.7 \text{ eV}$.

Na (2.75 eV) and K (2.30 eV) will show emission; Mo (4.17 eV) and Ni (5.15 eV) will not.

If the distance to the laser is changed, intensity changes but photon energy does not, so the result is unchanged – only Na and K will show emission.

Q.30 Light of intensity 10^{-5} W m^{-2} falls on a sodium photo-cell of a surface area of 2 cm^2 . Assuming that the top 5 layers of sodium absorb the incident energy, estimate the time required for photoelectric emission in

the wave picture of radiation. The work function for the metal is given to be about 2 eV. What is the implication of your answer?

Ans:

- Using the wave theory, the time for an electron to absorb enough energy is extremely large, showing a contradiction with experiment.
- Intensity $I = 10^{-5} \text{ W/m}^2$; area $= 2 \times 10^{-4} \text{ m}^2$; total absorbed power $P = 2 \times 10^{-9} \text{ W}$
- Total electrons (5 layers): $n' = 5 \times [(2 \times 10^{-4})/10^{-20}] = 10^{17}$
- Energy absorbed per electron per second: $E = P/n' = 2 \times 10^{-26} \text{ J/s}$
- Work function $\Phi_0 = 3.2 \times 10^{-19} \text{ J}$, so time $= \Phi_0/E = (3.2 \times 10^{-19})/(2 \times 10^{-26}) = 1.6 \times 10^7 \text{ s} \approx 0.5 \text{ years}$
- This is inconsistent with observed instant emission, showing that the wave theory cannot explain photoelectric effect.

Q. 31 Crystal diffraction experiments can be performed using X-rays or electrons accelerated through appropriate voltage. Which probe has greater energy? (For quantitative comparison, take the wavelength of the probe equal to $1 \text{ \AA}$, which is of the order of interatomic spacing in the lattice) ($m_e = 9.11 \times 10^{-31} \text{ kg}$).

Ans:

For the same wavelength, photons (X-rays) have much higher energy than electrons.

1. For electron: $K.E = \frac{h^2}{(2m_e\lambda^2)} = \frac{(6.64 \times 10^{-34})^2}{[2 \times 9.1 \times 10^{-31} \times (10^{-10})^2]} = 2.4 \times 10^{-17} \text{ J} = 149.375 \text{ eV}$
2. For X-ray: $E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34} \times 3 \times 10^8)}{(10^{-10} \times 1.6 \times 10^{-19})} = 12.375 \text{ keV}$

So, photons have much higher energy for the same wavelength.

Q.32 (a) Obtain the de Broglie wavelength of a neutron of kinetic energy 150 eV. As you have seen in question 11.31, an electron beam of this energy is suitable for crystal diffraction experiments. Would a neutron beam of the same energy be equally suitable? Explain. ($m_n = 1.675 \times 10^{-27} \text{ kg}$).

(b) Obtain the de Broglie wavelength associated with thermal neutrons at room temperature (27 °C). Also, explain why a fast neutron beam needs to be thermalised with the environment before it can be used for neutron diffraction experiments.

Ans:

Neutrons at 150 eV have much smaller de Broglie wavelength than X-rays, while thermal neutrons' wavelength matches atomic spacing.

$$(a) \lambda = \frac{h}{\sqrt{(2m_n K.E)}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 1.675 \times 10^{-27} \times 2.4 \times 10^{-17}}} = 2.327 \times 10^{-12} \text{ m (much smaller than atomic spacing)}$$

So, 150 eV neutrons are not good for crystal diffraction.

(b) For thermal neutrons (room temperature): $E = (3/2)kT$, $\lambda = h/\sqrt{(2m_n E)} = 1.447 \times 10^{-10} \text{ m}$ (comparable to atomic spacing), so fast neutrons need to be slowed down ("thermalised") for these experiments.

Q.33 An electron microscope uses electrons accelerated by a voltage of 50 kV. Determine the de Broglie wavelength associated with the electrons. If other factors (such as numerical aperture, etc.) are taken to be roughly the same, how does the resolving power of an electron microscope compare with that of an optical microscope which uses yellow light?

Ans:

Accelerated electrons have much shorter de Broglie wavelength than yellow light, so the electron microscope can see smaller details.

1. Voltage $V = 50,000 \text{ V}$; $m_e = 9.11 \times 10^{-31} \text{ kg}$; $h = 6.6 \times 10^{-34} \text{ Js}$
2. Wavelength $\lambda = h/\sqrt{(2m_e eV)} = 5.467 \times 10^{-12} \text{ m}$
3. Optical microscope (yellow light): $\lambda = 5.9 \times 10^{-7} \text{ m}$
4. The electron microscope's resolving power is about 10^5 times better (since resolving power $\propto 1/\lambda$).

Q.34 The wavelength of a probe is roughly a measure of the size of a structure that it can probe in some detail. The quark structure of protons and neutrons appears at the minute length scale of 10^{-15} m or less. This

structure was first probed in the early 1970s using high-energy electron beams produced by a linear accelerator at Stanford, USA. Guess what might have been the order of energy of these electron beams. (Rest mass energy of electron = 0.511 MeV.)

Ans:

We use the de Broglie relation to find the energy corresponding to a 10^{-15} m wavelength.

- $\lambda \approx 10^{-15}$ m; $p = h/\lambda = 6.6 \times 10^{-34}/10^{-15} = 6.6 \times 10^{-19}$ kg·m/s
- Relativistic energy $E \approx p \times c = 6.6 \times 10^{-19} \times 3 \times 10^8 = 1.98 \times 10^{-10}$ J
- In eV: $1.98 \times 10^{-10}/1.6 \times 10^{-19} = 12.375 \times 10^8$ eV

So, electron beams of order ~ 1 GeV are needed for such small scales.

Q.35 Find the typical de Broglie wavelength associated with a He atom in helium gas at room temperature (27 °C) and 1 atm pressure, and compare it with the mean separation between two atoms under these conditions.

Ans:

Calculate He atom's de Broglie wavelength and use the ideal gas law for mean separation, then compare results.

- $T = 300$ K; atomic mass = 4 u = 6.64×10^{-27} kg
- $\lambda = h/\sqrt{(3mkT)} = 0.7268 \times 10^{-10}$ m

- Mean separation: $r = [kT/P]^{1/3} = [1.38 \times 10^{-23} \times 300 / 1.01 \times 10^5]^{1/3} = 3.35 \times 10^{-9} \text{ m}$
- The mean separation is much greater than the de Broglie wavelength.

Q.36 Compute the typical de Broglie wavelength of an electron in metal at 27 °C and compare it with the mean separation between two electrons in a metal which is given to be about $2 \times 10^{-10} \text{ m}$. (Note: Exercises 11.35 and 11.36 reveal that while the wave-packets associated with gaseous molecules under ordinary conditions are non-overlapping, the electron wave-packets in a metal strongly overlap with one another. This suggests that whereas molecules in an ordinary gas can be distinguished apart, electrons in a metal cannot be distinguished apart from one another. This indistinguishability has many fundamental implications, which you will explore in more advanced Physics courses.)

Ans:

At room temperature, every electron's de Broglie wavelength is much larger than its mean separation, making them overlap strongly.

- $T = 300 \text{ K}$; $m = 9.11 \times 10^{-31} \text{ kg}$; $h = 6.6 \times 10^{-34} \text{ Js}$
- $\lambda = h/\sqrt{(3mkT)} \approx 6.2 \times 10^{-9} \text{ m}$
- This is much larger than the average electron separation, $2 \times 10^{-10} \text{ m}$.

Q.37 Answer the following questions:

(a) Quarks inside protons and neutrons are thought to carry fractional charges $[(+2/3)e; (-1/3)e]$. Why do they not show up in Millikan's oil-drop experiment?

(b) What is so special about the combination e/m ? Why do we not simply talk of e and m separately?

(c) Why should gases be insulators at ordinary pressures and start conducting at very low pressures?

(d) Every metal has a definite work function. Why do all photoelectrons not come out with the same energy if incident radiation is monochromatic? Why is there an energy distribution of photoelectrons?

(e) The energy and momentum of an electron are related to the frequency and wavelength of the associated matter wave by the relations:

$E = h\nu$, $p = h/\lambda$. But while the value of λ is physically significant, the value of ν (and therefore the value of the phase speed) has no physical significance. Why?

Ans:

Each sub-question is briefly answered below for clarity.

- (a) Quarks cannot be isolated individually, so only combinations with integer charge are observed in Millikan's experiment.
- (b) The ratio e/m determines the path of a particle in fields, so it is more useful than e or m alone in many physics equations.

- (c) At ordinary pressure, gases have too few ions to conduct; at low pressure, mean free path increases, enabling conduction via ionization.
- (d) Photoelectrons have different energies due to variations in their positions and interactions within the metal; hence, even with monochromatic light, energies are spread.
- (e) λ corresponds to observable properties (like diffraction), but v (frequency) for matter waves is related to phase velocity, which does not represent actual particle motion and has no direct physical meaning.